

Template for Computer Programming I Practical Recording

Date	:	20/02/2025
Problem Specification	:	The problem trying to find square root of given number.
Assumption	:	User input is valid.
Limitation	:	input $a \geq 0$
Input	:	user input the number n
Processing	:	calculate the square root of given number n using given format
Output	:	output square root of given number n in 6 digit decimal number
Algorithm	:	Step1: Start Step2: Prompt as "Enter the number a" Step3: Read the value for a from the user Step4: if ($a \geq 0$) is false proceed to step 12 Step5: Get the random value for x0 Step6: Declare variable x $\leftarrow x_0$ Step7: Repeat Step 8, Step 9 until ($x^2 - a > 0.00001 \parallel a - x^2 > 0.00001$) Step8: declare variable x1: $x_1 \leftarrow (x + a/x)/2$ Step9: $x \leftarrow x_1$ Step10: if ($x > 0$) is true Print x as a six-digit decimal value, then go to Step13 Step11: else Print ($x^2 - 1$) as a six-digit decimal value then go to Step13 Step12: Print as "wrong input, Input a positive number or 0" Step13: End the Programme.
Programme listing	:	Programme file attached
Test data and expected output	:	1. Test data : a=15 Expected output : 3.872983
Output obtained for test data	:	1. Test data : a=15 Obtained output : 3.872983
Discussion	:	Take an idea on how to use while loops and set the range correctly.
Conclusion	:	The process ensures an accurate estimate of the square root. Finally, the result is displayed as a six-decimal value, providing a precise output.